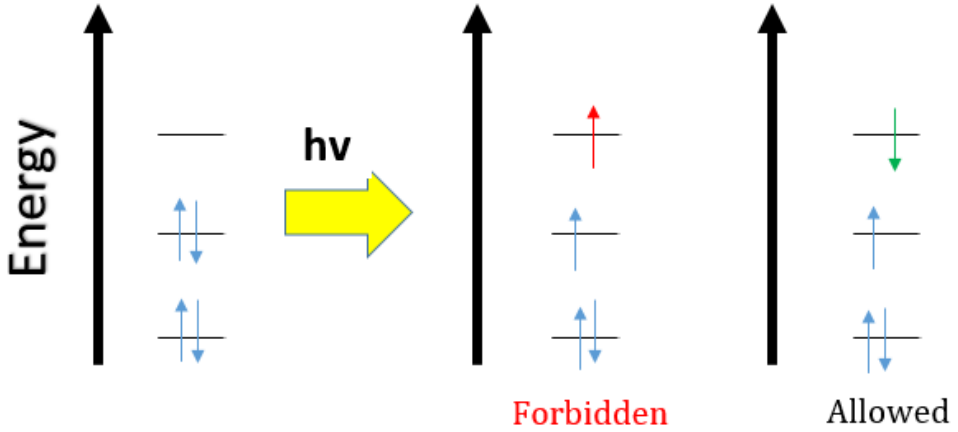
Selection Rules

<u>The Laporte rule</u>

In a <u>centrosymmetric environment</u> transitions between like atomic orbitals such as s-s, p-p, d-d or f-f transitions are forbidden.

It follows that **transitions in which the spin "direction" changes are forbidden**. **Formally**, <u>only states with the same total spin quantum number, are "spin-allowed"</u>.



Selection rule for electric dipole transitions

1. $\Delta l = \pm 1$ => Total orbital A.M Q. Number.
2. $\Delta m = 0, \pm 1$ => Magnetic Q. Number.
3. $\Delta S = 0$ => Total Spin Q Number
4. $\Delta J = 0, \pm 1$ => Total angular Momentum Q. Number

Region of the spectrum	Main interactions with matter
Radio	Collective oscillation of charge carriers in bulk material. An example would be the oscillation of the electrons in an antenna.
Microwave through far infrared	Molecular rotation
Near infrared	Molecular vibration
Visible	Molecular electron excitation
Ultraviolet	Excitation of molecular and atomic valence electrons
X-rays	Excitation and ejection of core atomic electrons, Compton scattering (for low atomic numbers)
Gamma rays	Energetic ejection of core electrons in heavy elements, Compton scattering (for all atomic numbers), excitation of atomic nuclei, including dissociation of nuclei
High energy gamma rays	Creation of particle-antiparticle pairs. At very high energies a single photon can create a shower of high energy particles and antiparticles upon interaction with matter.

Rotational Spectroscopy

Rotational spectroscopy or **microwave spectroscopy** studies the absorption and emission of electromagnetic radiation (typically in the microwave region of the electromagnetic spectrum) by molecules associated with a corresponding change in the rotational quantum number of the molecule.

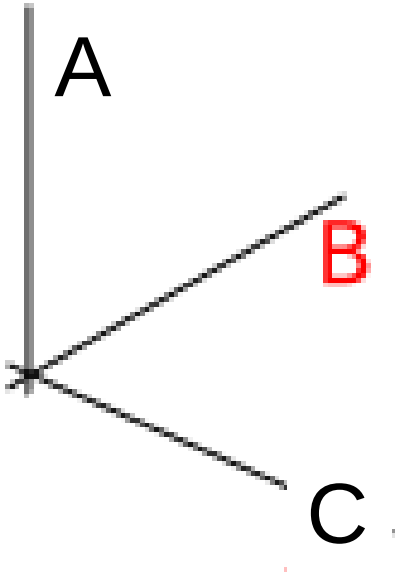
The rotation of a three-dimensional body

⬇

Resolve it into rotational components about three mutually perpendicular directions through the centre of gravity

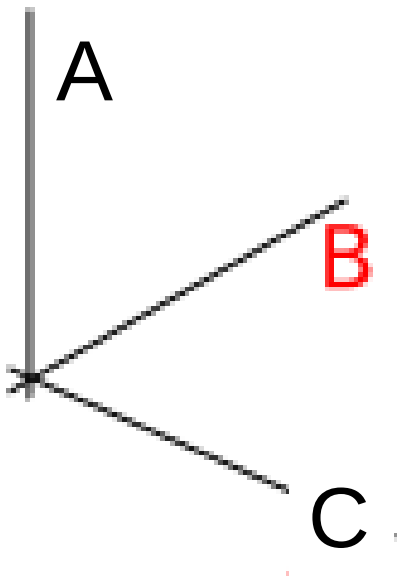
⬇

The principal axes of rotation



A rotating body has three principal <u>moments of inertia</u> ➡ one about each axis

Designated by
I_A, I_B, I_C



Moment of inertia (with respect to a given axis of rotation) of any object, be it a point mass or a 3D-structure, is given by:

$$I = \int r^2 \, dm$$

where *m* is mass and *r* is the perpendicular distance to the axis of rotation.

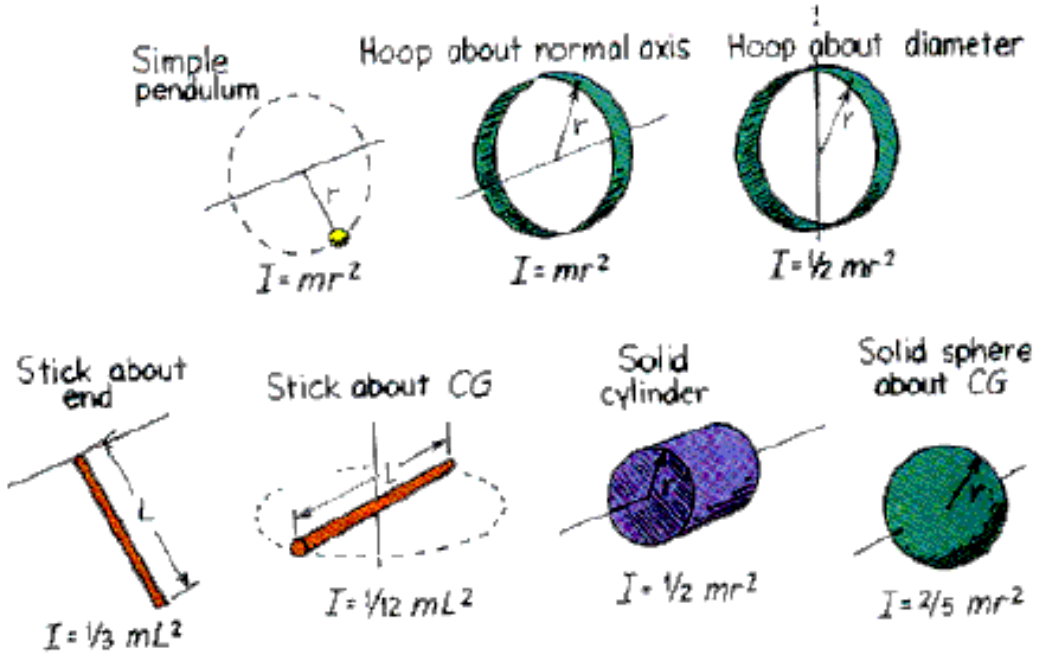
The **(scalar) moment of inertia** of a point mass rotating about a known axis is defined by

$$I = mr^2$$

The moment of inertia is additive.

For a rigid body consisting of N point masses m_i with distances r_i to the rotation axis, the total moment of inertia equals the sum of the point-mass moments of inertia:

$$I = \sum_{i=1}^{N} m_i r_i^2$$

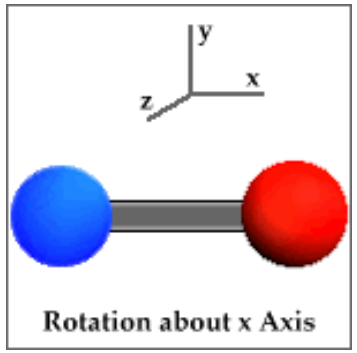


The mass distribution along the axis of rotation has no effect on the moment of inertia.

Linear molecules

All the atoms are arranged in a straight line.
Example: HCl

(a) Rotation about the bond axis ($I_A = 0$)



(b) End-over-end rotation in the plane of the paper

(c) End-over-end rotation at right angles to the plane

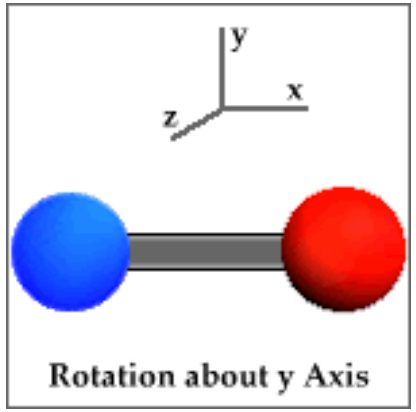
Directions of rotation:

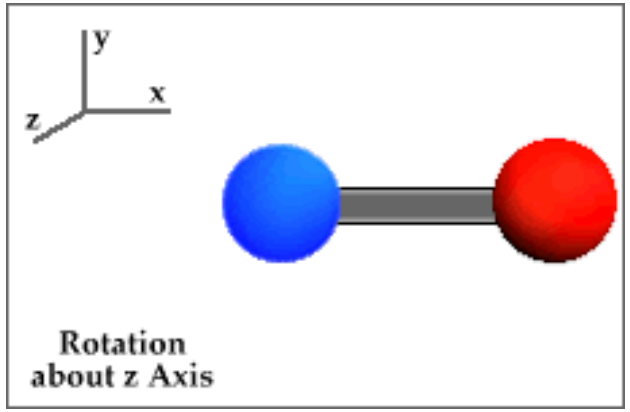
(a) About the bond axis (**$I_A = 0$**)

(b) End-over-end rotation in the plane of the paper

(c) End-over-end rotation at right angles to the plane

$$I_B = I_C$$

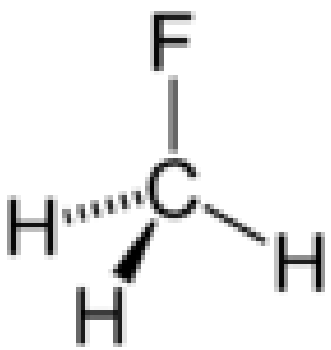




Symmetric tops

For symmetric top molecules

the end-over-end rotation in and out of the plane of the paper are identical. $I_B = I_C$

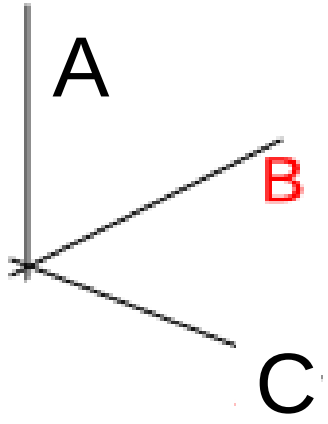


The C-F bond axis

The main rotational axis since the centre of gravity lies along it.

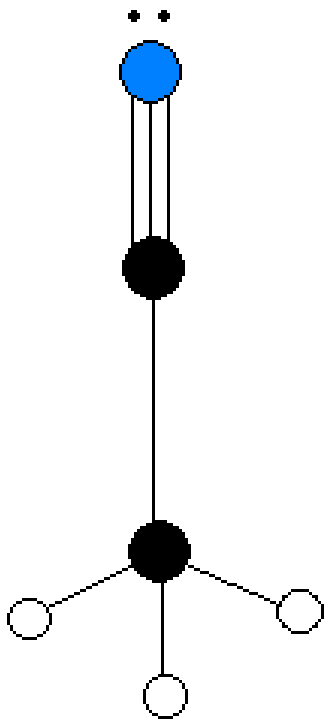
The moment of inertia is <u>not negligible</u> about this axis.

For symmetric tops:
$I_B = I_C \neq I_A$ and $I_A \neq 0$



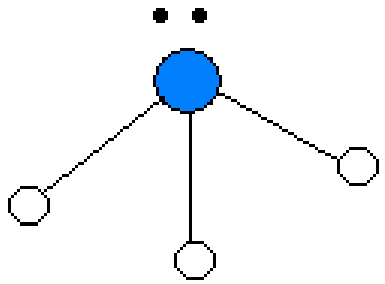
Oblate symmetric top

$I_B = I_C < I_A$



Prolate symmetric top

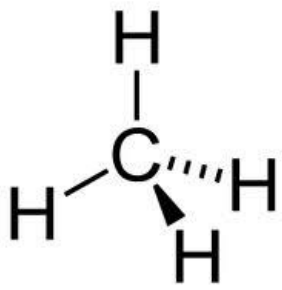
$I_B = I_C > I_A$



Spherical tops

All three moments of inertia are identical.

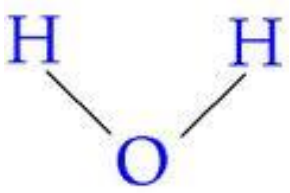
$I_A = I_B = I_C$



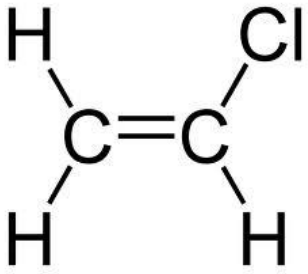
Asymmetric tops

All three moments of inertia are different.

$$I_A \neq I_B \neq I_C$$



water

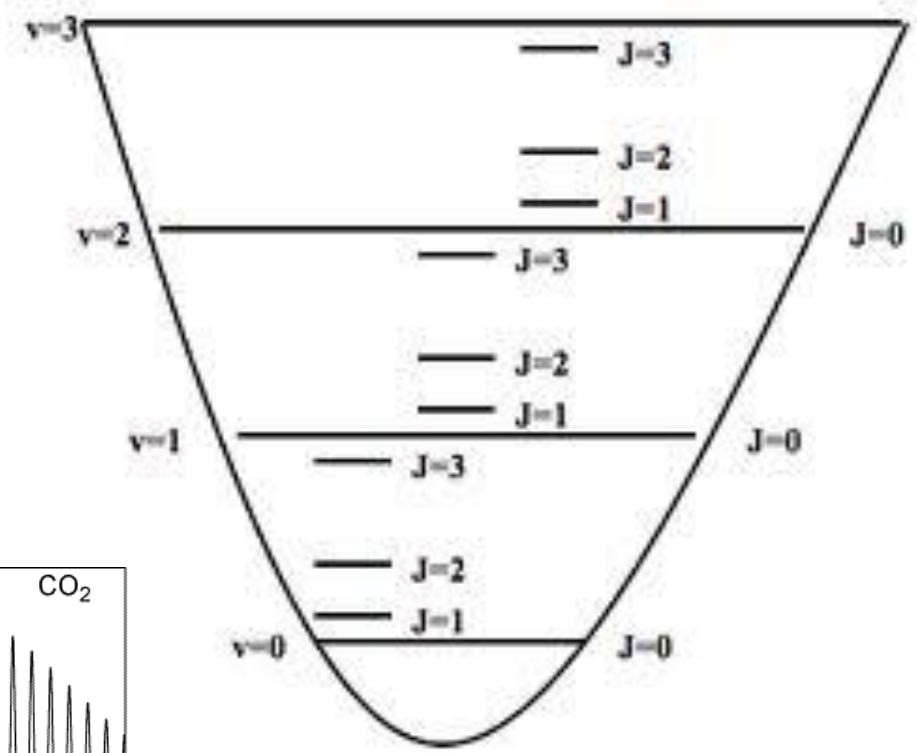


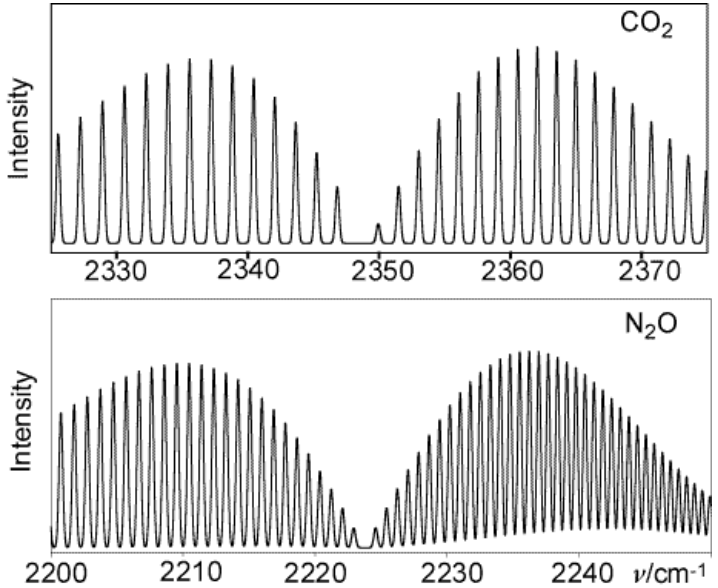
vinyl chloride

Description	Moments of Inertia	Examples
Monoatomic	$I_1 = I_2 = I_3 = 0$	Ar
Linear	$I_1 = 0; I_2 = I_3 > 0$	N_2, O_2, CO_2, N_2O
Spherical top	$I_1 = I_2 = I_3 > 0$	CH_4
Symmetric top	$I_1 \neq 0; I_2 = I_3 > 0$	NH_3, CH_3Cl, CF_3Cl
Asymmetric top	$I_1 \neq I_2 \neq I_3$	H_2O, O_3

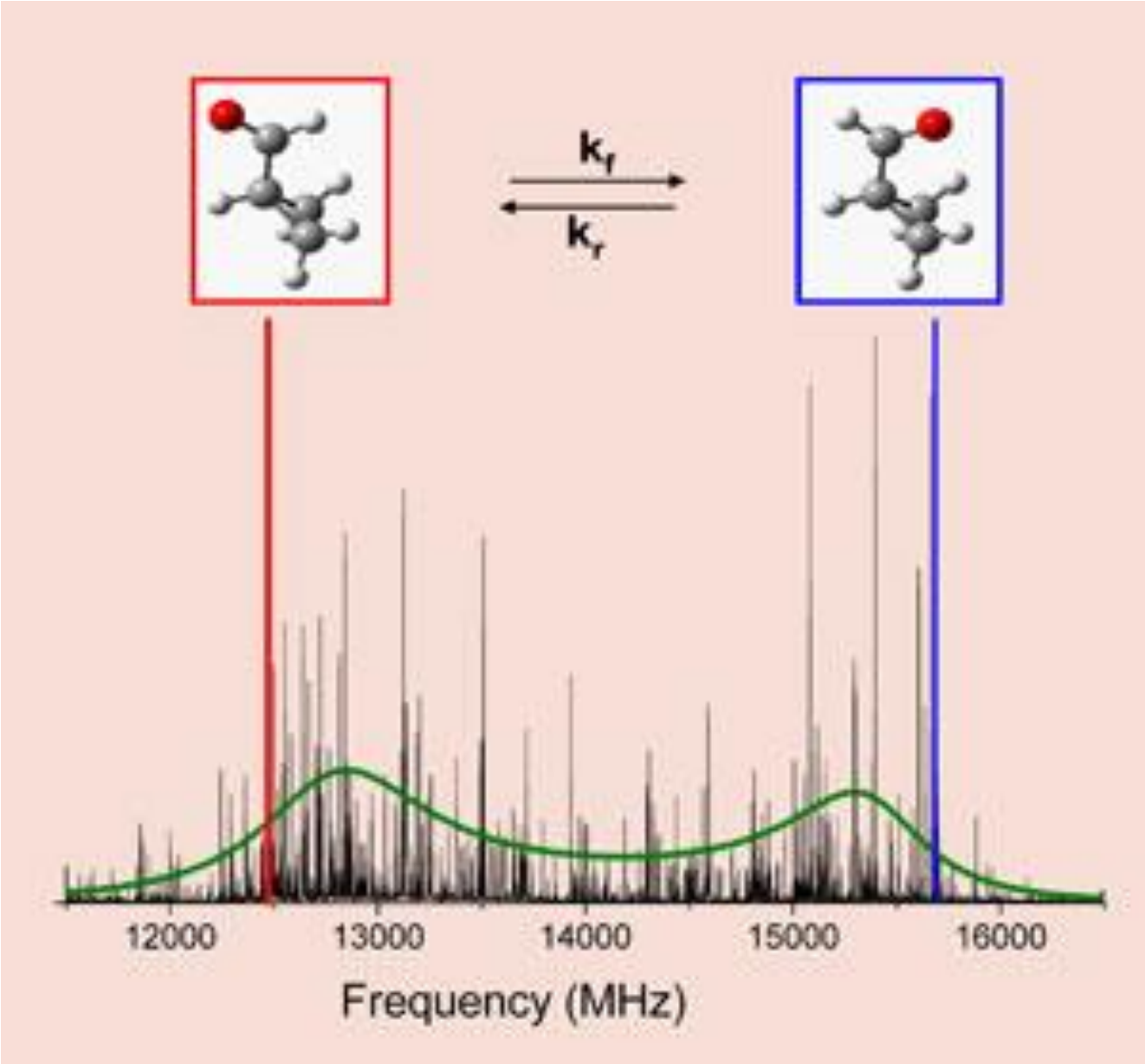
Rotational Spectra

- Rotational energy levels are quantized.
- The permitted energy levels are called rotational energy levels.
- Transition of molecules between these levels result into rotational spectra.





Isomerisation of cyclopropane carboxaldehyde



Diatomic molecules

The Rigid Diatomic molecule

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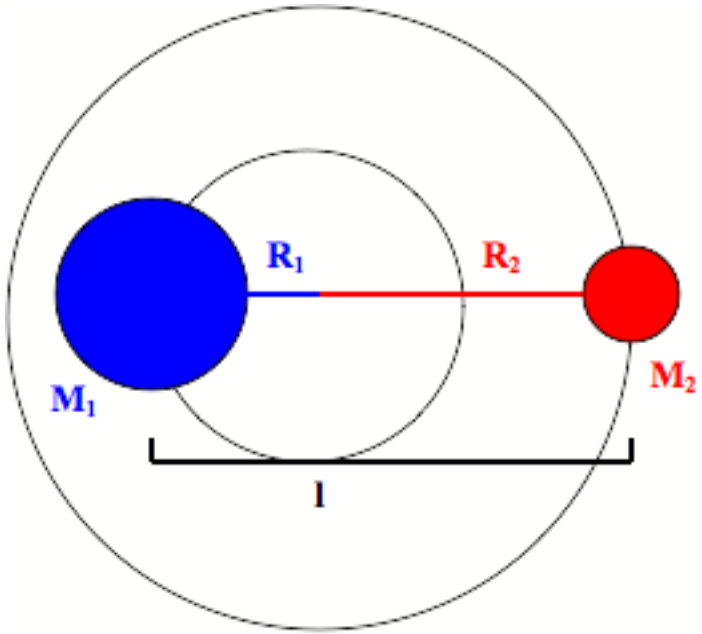
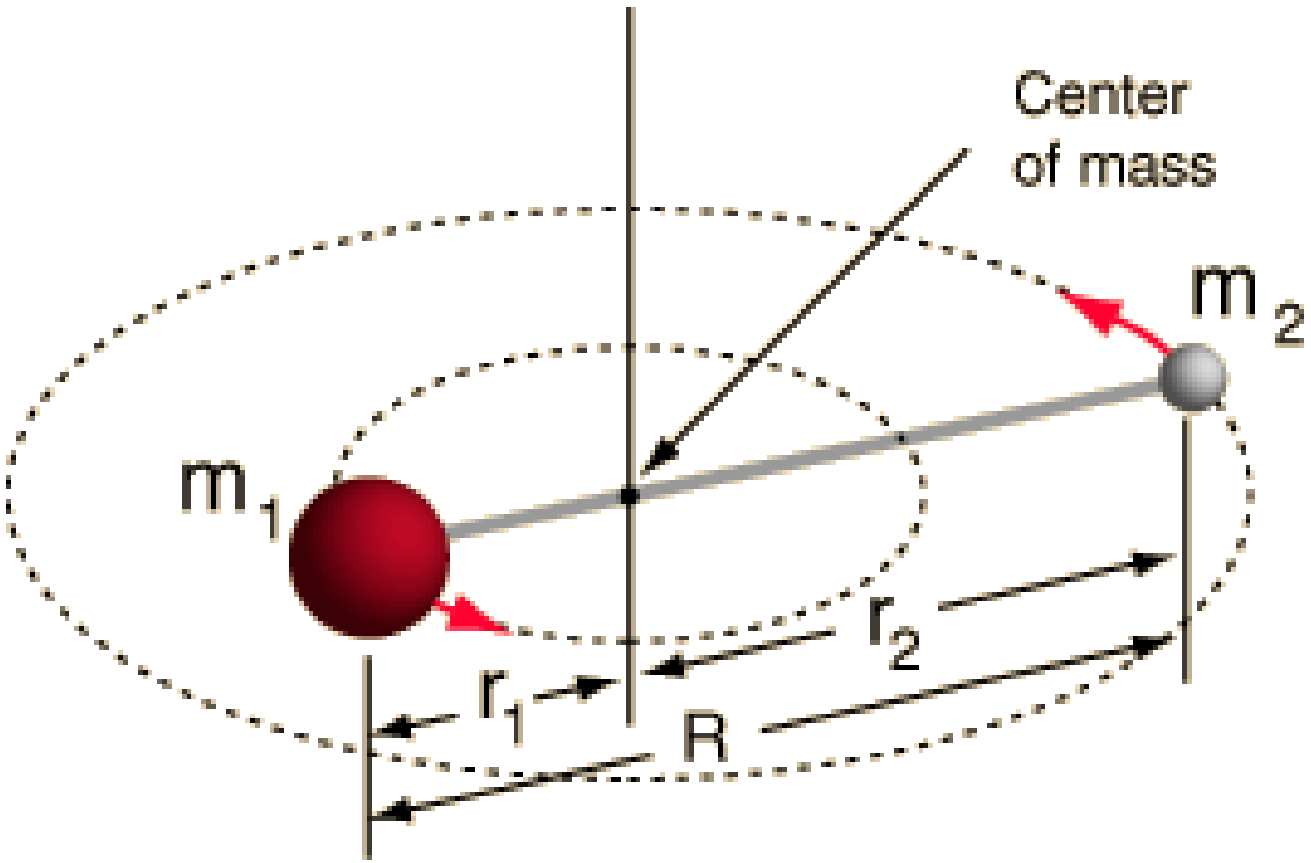
Simplest of all linear molecules





Masses m_1 and m_2 are joined by a rigid bar whose length is

$$\mathbf{R = r_1 + r_2}$$

The molecule rotates end-over-end about the centre of gravity: defined as the balancing equation: $\mathbf{m_1 r_1 = m_2 r_2}$

The moment of inertia about the centre of mass is defined by:

$$I = m_1 r_1^2 + m_2 r_2^2 \quad \text{But, } (m_1 r_1 = m_2 r_2)$$
$$= m_2 r_2 r_1 + m_1 r_1 r_2$$
$$= r_1 r_2 (m_1 + m_2)$$

Now, $m_1 r_1 = m_2 r_2 = m_2 (R - r_1)$

So, $$r_1 = \frac{m_2 R}{m_1 + m_2}$$ and $$r_2 = \frac{m_1 R}{m_1 + m_2}$$

$$I = r_1 r_2 (m_1 + m_2)$$

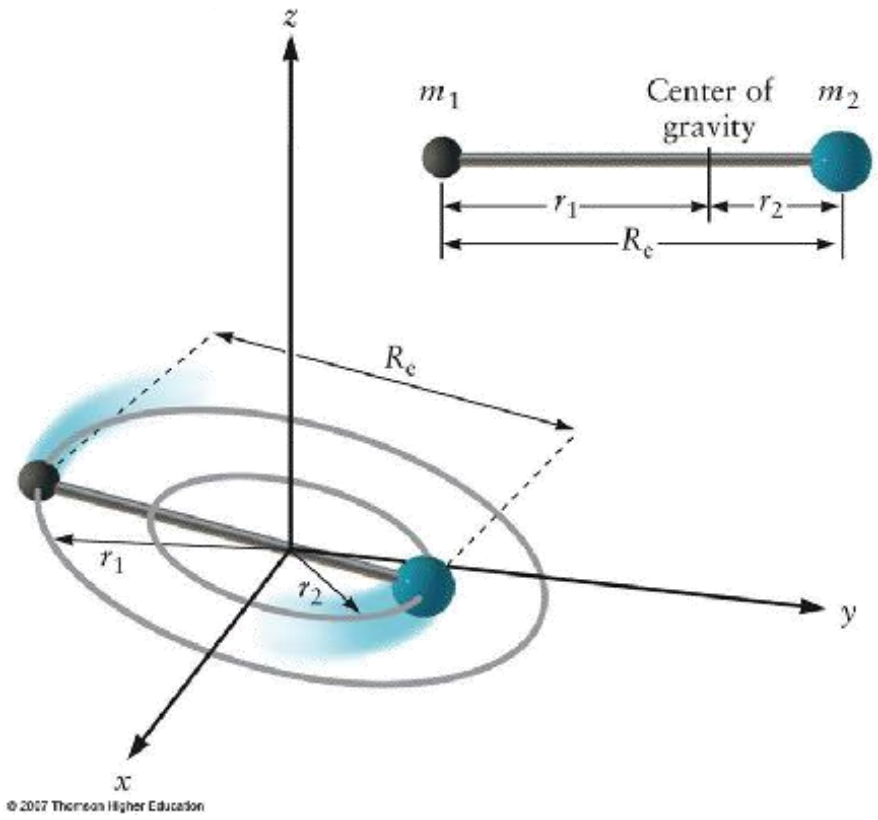
$$r_1 = \frac{m_2 R}{m_1 + m_2}$$

$$r_2 = \frac{m_1 R}{m_1 + m_2}$$

$$I = \frac{m_1 m_2}{m_1 + m_2} R^2 = \mu R^2$$

$$\mu = \frac{m_1 m_2}{m_1 + m_2}$$

Reduced mass of the system



If the molecule rotates in three dimensions, then we need two angles θ and ϕ to characterize it.

- the **angles in the spherical polar coordinate system** (θ is the polar angle and ϕ is the azimuthal angle).